

Integrated scientific decision: NO_PHYSICS_INFORMED_FAMILY_ADDS_MATERIAL_VALUE

Primary-field noninferiority	FAIL
Material predictive gain	PASS
Material physical gain	PASS
Functional graph benefit	PASS
S11 five-seed stability	NOT RUN
Full-state gate	FAIL / LIMITED
New external FEM panel	NOT RUN / OUT OF SCOPE
Sensor validation	NOT RUN / OUT OF SCOPE

R4_LHS_03 — diagnostic; not accepted

R4 passes material predictive, physical and frozen-checkpoint graph diagnostics, but fails primary-field noninferiority; therefore S11 five-seed confirmation was not run and no family is accepted. External FEM and sensors remain out of scope.